Building Convolutional Neural Networks for CIFAR-10 Image Classification

Abstract

This report presents the design, implementation, and evaluation of convolutional neural networks for image classification on the CIFAR-10 dataset. A baseline CNN and an enhanced MiniVGG-style CNN were developed and trained using regularization, data augmentation, and learning-rate scheduling. The models were evaluated using accuracy, confusion matrices, per-class performance, misclassification analysis, and GradCAM visualizations to better understand model behaviour.citewrite.qut.edu+2

1. Introduction

The objective of this assignment was to design and train CNN models for classifying CIFAR-10 images into 10 categories: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. CIFAR-10 is a standard benchmark for computer vision because it contains small color images with enough variation to make classification challenging while still being manageable for experimentation.adobe+1

The work focused on comparing a simple baseline network with a deeper architecture, then improving performance using augmentation, dropout, batch normalization, and learning-rate scheduling. In addition, interpretability techniques such as GradCAM were used to understand where the model focused while making predictions.sussex.ac+1

2. Dataset and Preprocessing

The CIFAR-10 dataset contains 60,000 color images of size 32×32 across 10 classes, with 50,000 training images and 10,000 test images. For training, the images were normalized using CIFAR-10 mean and standard deviation values to stabilize optimization and improve convergence.chrmp+1

The data pipeline also included augmentation to improve generalization. Random horizontal flips, random rotations, and random crops with padding were applied during training. These transformations help the model learn more robust features and reduce overfitting to the training set.clickhelp+1

3. Model Architectures

3.1 Baseline CNN

The baseline model was intentionally simple and followed the required assignment structure. It used two convolutional blocks followed by a fully connected classifier:

- **Conv2D(32, 3×3) → ReLU → MaxPool(2×2)**
- **Conv2D(64, 3×3) → ReLU → MaxPool(2×2)**
- **Flatten → Dense(128) → ReLU → Dropout(0.5) → Dense(10)**

This architecture provides a strong starting point while remaining computationally efficient. However, because it is shallow, it has limited capacity for learning complex hierarchical image features.

3.2 MiniVGG Model

The advanced model was a MiniVGG-style network with three convolutional blocks, batch normalization, and dropout. Each block used multiple convolution layers followed by max pooling. This deeper architecture improves representational power and generally performs better on CIFAR-10 than a small baseline CNN.

Batch normalization was added to stabilize gradients and speed up training. Dropout was used in the classifier to reduce overfitting. Compared with the baseline model, MiniVGG is more expressive and better suited for learning subtle patterns in small images.sussex.ac+1

4. Training Procedure

The models were trained using categorical cross-entropy loss. The baseline network used SGD with momentum, while the advanced model used Adam with weight decay. A learning-rate scheduler based on validation loss was included to reduce the learning rate when improvement slowed down.

Early stopping was used to prevent overtraining, and the best validation checkpoint was saved during training. This approach is standard in technical model development because it helps balance accuracy, stability, and generalization.adobe+2

5. Results and Evaluation

The training and validation curves showed that the baseline CNN learned the task but converged more slowly and achieved lower accuracy than the advanced model. The MiniVGG architecture trained more effectively and produced better generalization on the validation and test sets.

A confusion matrix was generated for the test set to identify which classes were often confused. As expected, visually similar classes such as cat and dog, or automobile and truck, were more likely to be misclassified. These errors reflect the limited resolution of CIFAR-10 images and the overlap in object shape, texture, and

background context.chrmp+1

Per-class accuracy analysis confirmed that some classes were easier than others. Classes with more distinctive shapes, such as ship or airplane, were generally predicted more accurately than classes with finer visual similarity.

6. Error Analysis

The misclassified image grid showed several common failure patterns. First, small objects occupying a limited region of the image were harder to classify. Second, images with cluttered or ambiguous backgrounds caused incorrect predictions. Third, classes with similar structure or texture were frequently confused.

This suggests that model performance is limited not only by architecture but also by the intrinsic difficulty of CIFAR-10. A deeper model helps, but there are still ambiguous examples where even humans may hesitate.

7. GradCAM Interpretation

GradCAM heatmaps were generated to visualize which regions influenced the model's decisions. For correct predictions, the highlighted regions often aligned with the object of interest, showing that the model had learned useful spatial attention. For incorrect predictions, the heatmaps sometimes focused on background regions or irrelevant textures.

This interpretability step is important because it shows that high accuracy alone is not enough. A model should also make decisions for meaningful reasons, especially when used in real applications.grammarly+1

8. Advanced Experiments

Transfer learning with a pretrained model such as ResNet18 is expected to improve performance further because the model starts with useful feature representations learned from a large dataset. Test-time augmentation can also improve accuracy by averaging predictions across slightly altered versions of the same image.

These advanced methods usually outperform training from scratch, especially when the dataset is relatively small or when the task requires stronger generalization. They also demonstrate the trade-off between model complexity and practical performance.clickhelp+2

9. Discussion

Data augmentation improved generalization by exposing the model to more varied versions of the same training images. Batch normalization made training more stable and reduced sensitivity to initialization. Dropout helped control overfitting, especially in the fully connected layers.

The learning-rate scheduler improved convergence by lowering the step size when validation loss stopped improving. This allowed the optimizer to continue making smaller refinements instead of oscillating around a local minimum. Overall, the advanced model achieved better results because it combined deeper feature extraction with regularization and more careful training control.

10. Conclusion

This assignment demonstrated the full deep-learning workflow for image classification: data preparation, model design, training, evaluation, and interpretation. The baseline CNN provided a simple reference point, while the MiniVGG model improved performance through depth, batch normalization, dropout, and learning-rate scheduling.

The results show that model quality depends not only on architecture but also on preprocessing, augmentation, and training strategy. GradCAM and confusion-matrix analysis added interpretability, making the project more complete and useful for understanding how CNNs behave on real image data.sussex.ac+2

11. References

1. **CIFAR-10 dataset description and benchmark details.adobe+1**
2. **Technical report writing structure and formatting guidance.citewrite.qut.edu+2**
